S1Q05

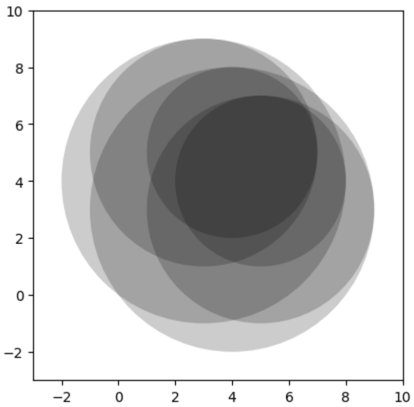
Question 5 - Stage 1

Find the area (to 3.d.p.) of the region made up of all the points that are inside all six circles of the form $(x-a)^2+(y-b)^2=c^2$ where $\{a,b,c\}=\{3,4,5\}$.

Working

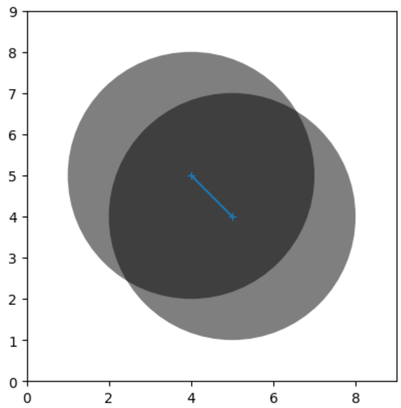
Plot All Circles

In [1]:
```
from itertools import permutations
import matplotlib.pyplot as plt

numbers = [3,4,5]
fix,ax = plt.subplots()

for circleArrangement in permutations(numbers,len(numbers)):
    center = (circleArrangement[0],circleArrangement[1])
    radius = circleArrangement[2]

    circle = plt.Circle(center,radius)
    circle.set_facecolor((0,0,0,0.2))
    ax.add_patch(circle)


ax.set_aspect('equal',adjustable='box')
ax.set_xlim(-3,10)
ax.set_ylim(-3,10)
plt.show()
```

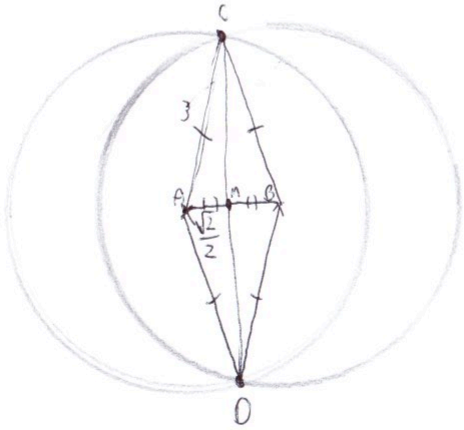


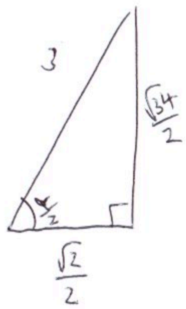
Plot only the two relevant ones

In [2]:
```
import matplotlib.pyplot as plt


fix,ax = plt.subplots()

center0 = (4,5)
center1 = (5,4)
radius = 3

circle0 = plt.Circle(center0,radius)
circle0.set_facecolor((0,0,0,0.5))

circle1 = plt.Circle(center1,radius)
circle1.set_facecolor((0,0,0,0.5))

ax.add_patch(circle0)
ax.add_patch(circle1)

ax.plot([center0[0],center1[0]],[center0[1],center1[1]],marker="+",)

ax.set_aspect('equal',adjustable='box')
ax.set_xlim(0,9)
ax.set_ylim(0,9)
plt.show()
```



Final Calculations





$$\cos\left(\frac{\alpha}{2}\right) = \frac{\sqrt{2}}{6}$$

$$\frac{\alpha}{2} = \arccos\left(\frac{\sqrt{2}}{6}\right)$$

$$\alpha = 2\arccos\left(\frac{\sqrt{2}}{6}\right)$$

$$|\vec{AB}| = \sqrt{2}$$

$$|\vec{AC}| = |\vec{BC}| = 3$$

$$\angle CAD = \angle CBD = 2\arccos\left(\frac{\sqrt{2}}{6}\right)$$

$$\frac{\text{Area}}{2} = \frac{\alpha}{2\cancel{\pi}}\cancel{\pi} r^2 - \frac{1}{2}\sqrt{34}\cdot\frac{\sqrt{2}}{2}$$

$$= \frac{\alpha}{2} r^2 - \frac{1}{2}\sqrt{17}$$

$$\text{Area} = \alpha r^2 - \sqrt{17}$$

$$= 18\arccos\left(\frac{\sqrt{2}}{6}\right) - \sqrt{17}$$

$$\approx 19.8682\ldots$$

$$\simeq 19.868$$